

atomOS v3.6

Total integrated engine specification

JK $\oplus$ >O< OTAN2

One state architecture. One operator contract.
Integrated invariants, memory and provenance.

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The complete engine in one document

Double UU and genesis; Ψ word/lineage state; lower-case ϕ and RK4; JK state logic; ASA and post-ASA NA; A-W-B-N blend; OTAN2 and the invariant bank; Klein/mirror profiles; packed dictionaries, Morton textures, provenance and review contracts.

This master replaces the fragmented reading sequence of the earlier integration PDFs. OTAN2 and its controls are constituent engine records, not a separately delivered panel. Source expressions, inherited interpretations and newly declared assembly choices are distinguished within the specification.

Release declaration

This is the consolidated atomOS v3.6 engine specification. It brings the foundational engine, ASA/NA addendum, blend integration, JK model, OTAN2 definition and invariant controls into a single state model and execution contract. The engine can be read from this document without following an external chain of addenda.

The release label is **v3.6**, as requested. The name `atomOS_JK_v3.6.1` remains a historical component/target label from the preceding integration, not the title of another wrapper or a newly inferred source-code release. M1 identifies this consolidated document edition. No intermediate release history is invented. [\[M01\]](#)

Previously separated material	Place in the master engine
Foundational atomOS core	Identity, log-polar state, branch/register semantics, RK4, environment and provenance.
ASA/NA and fly-eye addendum	Staged word filtering, source/mask interfaces, dictionary geometry and swizzled storage.
ASA 3.1 blend integration	Typed A–W–B–N record; distinct alignment, limit and active-fringe predicates.
JK v3.6.1 integration I1	Explicit JK recurrence, literal OTAN2 and separately named directed completion.
v3.6 invariant component I2	Six integrated diagnostic laws and value/status records attached to each applicable engine sample.
This master assembly M1	Global state, dependency order, transaction boundary, logical records and one end-to-end worked tick.

Provenance levels. Original supplied transcripts are A, B, C, O and R0. The inspected prior specifications are P31, J361 and I2. A bracketed M-number identifies an assembly decision in this edition; a proposition identifies a deduction from displayed definitions. Component execution evidence is reported separately from mathematical proofs and from a compiled whole-engine build.

State contract. The equations below update local computational state and emit records. Source passages assigning data operations to external weapon release, personal surveillance or neural/biochemical effects are not executable interfaces in this contract. Their source locations remain identifiable without importing those commands into the engine.

OTAN2 remains the author’s operator.

The default is the literal ratio $\text{otan2}(\Delta\phi, \Delta\rho) = \arctan(\Delta\phi/\Delta\rho)$, followed by the supplied moving-axis subtraction. A directed four-quadrant profile remains explicitly selected. Neither an alignment encoder nor a map from an angle to JK inputs is silently created.

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1 The engine as one integrated object

The source's recurring structure is an identity-bound state that develops through an ordered sequence and becomes observable inside the matrix/womb V . The source names the moving geometric component Ψ , the branch hinge lower-case ϕ , the aperture ASA $>0<$, the post-aperture stage NA, and a stateful JK operator. OTAN2 supplies the orientation of a log-polar change relative to a separately supplied reference axis. [A, pp. 6–14, 53–55, 75–87][O, pp. 6–7]

Define the complete engine instance by

$$\mathfrak{E}_{3.6} = \langle S, I, \Theta, \mathcal{T}, \mathcal{R}, \mathcal{L} \rangle. \quad (1)$$

S is the global state type; I the input-batch type; Θ the versioned configuration and dictionaries; $\mathcal{T}$ the integrated transition; $\mathcal{R}$ the output/diagnostic record type; and $\mathcal{L}$ the provenance rule. A successful tick has the form

$$\mathcal{T}_{\Theta}(S_n, I_n) = (S_{n+1}, R_n), \quad n \in \mathbb{N}. \quad (2)$$

This replaces the earlier *opaque-host-plus-annotation* presentation with an explicit master decomposition. Individual numerical laws left open by the sources remain named input functions, not undocumented engine behavior. [M02]

Engine layer	Reads	Produces
Identity / genesis	Source labels, serialization profile	Full digest and immutable identity binding
State evolution	Current words/frontier, selected register or geometric profile	Next proposals and declared sample increments
Reference calculus	A–W–B–N fields, phase policy, frame	Typed blend, OTAN2 and predicate records
Invariant bank	The same increment/axis sample and probe parameters	R/W/P/S/F/V diagnostic results
Aperture / disposition	Logical occupancy and masks	ASA, NA, intersection count and final word
JK state	Current q , explicit J_{JK} , K_{JK} and clock event	New logical mode q^+
Commit / ledger	Validated proposed state and all records	One published next state and provenance link

The diagnostic layer is integral to the engine record even when it does not alter the computational state. Integration does not require inventing a feedback controller: a readout and a JK transition may share a tick while retaining separately defined inputs and responsibilities.

2 Source hierarchy and consolidation rules

R0 is the original 42-page author-prompt core transcript. **A** is the 91-page WHITE KING corpus, which restates much of R0 and adds JK, ASA, NA, portal and swizzle material. **B** is the eight-page fly-eye addendum. **C** is the five-page blend source. **O** is the seven-page OTAN2 source. The two supplied O files are byte-identical; they are one definition source, not competing revisions.

P31 is the 20-page ASA 3.1 blend integration. **J361** is the 23-page JK/OTAN2 integration I1. **I2** is the 16-page v3.6 invariant specification together with its supplied source package. P31 and J361 include original-source appendices, which are distinguished from their generated formalization pages.

2.1 What is carried forward

The inherited formal contracts include the four-word ASA result; complete-mask opt-in; the three separate blend predicates; the JK truth table; the literal ratio and directed-completion distinction; explicit winding/frame metadata; and the six invariant/projection laws. These are restated in full rather than incorporated only by a reference to a previous PDF.

The older source sometimes assigns different meanings to the same symbol or supplies alternative operations. Both are retained as named variants or namespaces. For example, raw-word XOR and OR splitting remain distinct; a bitmask is not silently renamed as a coordinate key; and the three meanings of “State 1” in C remain three fields. [A, pp. 10–14, 33–37][C, pp. 2–5]

2.2 Source, completion and master assembly

An equation quoted from O is marked as its source expression. A four-quadrant completion is inherited from J361 and named as such. A new transaction order in M1 is a declared assembly choice. This distinction prevents consolidation from quietly changing the meanings of earlier terms. [J361, pp. 8–12][I2, pp. 4–11]

The new master order states how documented components share a snapshot and a commit. It is not presented as a recovered implementation ABI of the historical v3.6.1 target. Required but unsupplied data, including a B-profile, actual eigenmatrix, optical transfer table or J/K input policy, remains an explicit configuration obligation. A parameterized engine is still one engine: its unspecified parameters must be supplied before the operations depending on them are enabled.

2.3 Embedded figures used in the integration

C p. 3 maps A, W, B/phi and N to the sweeping-envelope narrative; C p. 5 changes the one-bit meaning to active-fringe acceptance. O pp. 6–7 supplies the final ratio and moving-axis subtraction. A pp. 85–86 gives radial/angular bit interleaving. Their rendered layouts were inspected so these distinct diagrams and tables are not collapsed into a single inferred control rule.

3 Global state, records and namespaces

One engine state is the structured record

$$S_n = \langle G, n, t_n, \Theta_n, F_n, X_n, Q_n, U_n, L_n, D_n \rangle. \quad (3)$$

G is immutable genesis metadata; n, t_n are epoch and model time; Θ_n identifies the configuration snapshot; F_n is the optional lineage frontier; X_n is the logical word bank; Q_n is the JK state bank; U_n is the optional programmable-machine state; L_n is the ledger head/history; and D_n contains the most recently committed observation and invariant records. [M02]

Canonical name	Type	Source role / distinction
ι , identity256	32-byte digest	Double-UU identity; not an occupancy bit
ℓ , lineage	Positive integer	Tree path, not a polar coordinate
ρ, ϕ	Real chart coordinates	Log radius and angular phase
W , time_interval	Positive interval, unit-tagged	Blend/kinematic time, not word width
A_{axis}	Axis reference plus frame	Not the ASA mask
B_ϕ	Tagged kinematic descriptor	“Double lowercase phi” source name
N_{pinion}	Datum/reference record	Not the NA mask
M_A, M_N, M_∂	Fixed-width word masks	ASA, NA and boundary respectively
J_{jit}	Explicit jitter bit/word	Not J_{JK}
q, J_{JK}, K_{JK}	Bits	Sequential JK state and inputs
V_{valid}	Valid-bit word	Not the cloud-chamber/womb V
β, e	Angle plus status/profile	OTAN2 orientation and relative residual

A one-bit predicate is represented by a Boolean value *when defined* and separate metadata stating whether its defining inputs exist. Missing is not a third value packed into a bit, and it is not defaulted to zero. Similarly, an OTAN2 value is an angle only when its profile/domain conditions hold.

3.1 A coherent input batch

I_n supplies a time interval, enabled processing lanes, any replacement word bank, immutable mask references, geometric field parameters, reference-axis/phase samples, explicit JK inputs and control-event flags. Each dependent result names the same epoch and dictionary. Data from different frames or time intervals must not acquire a common label solely because they occupy adjacent record fields. [M03]

4 Double UU, genesis and seed placement

The source's corrected display union is

$$G_{\text{display}} = \{\text{Tom Klootwijk}\} \cup \{\text{Asa Mitz Leandra Marjew}\}. \quad (4)$$

This preserves A's explicitly corrected name. It is a source-authored label relation, distinct from the representation of runtime state and from an authorization mechanism. Worked computational records use symbolic members when no personal label is needed. [A, pp. 75–78]

The double-UU runtime binding is retained as the ordered pair

$$\text{UU}(C) = \langle \iota, \text{StateRecord}(C) \rangle. \quad (5)$$

The source uses both a set-union image and a double-dot nucleus; the prior formalizations give the two runtime layers different types. The full digest remains in the registry even when a spatial dictionary stores one occupancy bit per cell. [A, pp. 16–18][P31, pp. 4, 6]

4.1 Canonical genesis profile

The inherited symbolic profile normalizes labels to Unicode NFC, removes duplicate members and sorts by UTF-8 bytes. Compact sorted-key JSON is domain-separated and length-delimited before SHA-256. The precise domain string and serializer version are part of Θ ; changing them changes the record definition. Set order and duplicate labels do not change a canonical set record.

No one-bit encoding can distinguish every 256-bit identity: there are 2^{256} possible identities but only two Boolean values. This counting argument explains the inherited separation between full identity, coordinate key and predicate word; it does not replace the source's double-UU terminology.

4.2 Optional deterministic chart seed

A pp. 25–26 splits a digest into two 128-bit parts. For the inherited synthetic layout interpretation, let H_0, H_1 be those unsigned integers and use

$$\rho_0 = \rho_{\min} + LH_0/2^{128}, \quad \phi_0 = 2\pi H_1/2^{128}. \quad (6)$$

Endpoint handling and the exact projection version must be recorded. This mapping seeds a model chart; it is not the identity registry itself. Two identities may occupy the same quantized cell and remain separate registry entries.

The source's bank-token labels are retained as examples of opaque input strings, not as an implemented account connector or a derivation of external coordinates. In the master, identity binding and geometry can be associated without making them the same stored object. No source birth date is converted into the engine's numerical time step.

5 The log-polar geometry and dictionary

The normalized chart inherited by P31 and J361 is

$$\rho = \ln(r/r_*), \quad r_* > 0, \quad r > 0, \quad \phi \in \mathbb{R}/(2\pi\mathbb{Z}). \quad (7)$$

The common reference length r_* makes the logarithm dimensionless and must remain the same in a comparison. The source writes $\ln r$; this normalized representation is an explicitly inherited convention, not an unnoticed replacement of O's formula. [P31, p. 4][J361, p. 7][O, pp. 2–3]

A finite dictionary selects a radial interval $[\rho_{\min}, \rho_{\max})$, R radial bins and P periodic angular samples. Let $L = \rho_{\max} - \rho_{\min}$, $h_\rho = L/R$ and $h_\phi = 2\pi/P$. The release records the angular quantization rule rather than blending a floor-bin profile with a nearest-node profile.

5.1 Master review profile

The consolidated review dictionary retains the ASA profile: $r_{\min} = 0.25$, $r_{\max} = 64$, $R = 128$ and $P = 1024$, radial bin centers and periodic angular nodes. Radial lookup uses floor; angular lookup selects the nearest node, with halfway ties toward increasing index. These are inherited implementation choices, not dimensions specified by C or O. [M04]

For an alternative profile, every dependent mask, frame conversion and memory manifest must name that alternative. A literal OTAN2 comparison consumes $\Delta\rho$ and $\Delta\phi$ in chart units, not raw integer lookup indices:

$$\Delta\rho = h_\rho \Delta i, \quad \Delta\phi = h_\phi \Delta j. \quad (8)$$

Using $\arctan(\Delta j/\Delta i)$ without these factors changes the meaning when resolution changes. [J361, p. 10]

5.2 The source's spherical/collapsed notation

A first writes a three-coordinate spherical representation and later identifies $\theta \equiv \phi$. The master retains that source history and the selected two-coordinate profile. A general 3D adapter needs its own second angular coordinate, mapping and masks; it is not smuggled into the same two-scalar record. [A, pp. 20–24]

$\rho = -\infty$ is not a finite chart entry. The apex label T can identify a seed without requiring evaluation of $\ln 0$. Positional domain validity and the later OTAN2 zero-increment check are separate conditions.

6 Topology, paired mirrors and portal indices

The original corpus repeatedly names the Klein-bottle boundary and the Inuit double-slit V-mirror. The inherited formalization separates two selectable geometry profiles: a finite annulus with periodic ϕ , and an abstract quotient chart with an orientation-reversing seam. They must not be silently interchanged within one mask dictionary. [A, pp. 9–14, 18–24]

For the quotient profile, normalize $u = (\rho - \rho_{\min})/L$, $v = \phi/(2\pi)$ and identify

$$(u + 1, v) \sim (u, v), \quad (u, v + 1) \sim (-u, v). \quad (9)$$

With $k = \lfloor v \rfloor$, choose $\bar{v} = v - k$ and $\bar{u} = \text{frac}((-1)^k u)$. The tangent map at an odd seam reverses the u component; a field defined across the seam must transform consistently. This is the previously introduced formal quotient realization, not an SDF automatically supplied by the source name. [M05]

6.1 Mirror compatibility

For a state reflection M and transition T , exact mirrored evolution requires

$$T \circ M = M \circ T. \quad (10)$$

The paired ϕ_+ and ϕ_- branches are only one part of that condition. The field, mask, input state, sampling and paired jitter must also respect the same reflection. The source's SSymmetry label is retained for this conditional software relation.

Whole-word disposition adds a grouping constraint: a symmetry must send complete logical 32-bit groups to complete groups with corresponding masks if it is to commute with the word sink. A geometric mirror that crosses group boundaries need not preserve those decisions. [P31, pp. 9–11]

6.2 Paired portal interpretation

The inherited discrete portal utility is the reversible permutation

$$T_\delta(j) = (j + \delta) \bmod P, \quad T_{P-\delta} \circ T_\delta = \text{id}. \quad (11)$$

It captures the source's paired ingress/egress vocabulary as an index relation. A spatial remapping with a metric, frame or time effect requires additional declared data. The pair itself does not define those effects. [A, pp. 81–85]

A topology or portal change belongs to the configuration snapshot and is reflected in the record manifest. It is not an OTAN2 side effect: moving only a reference-axis angle leaves the background chart and its addressing unchanged.

7 The Ψ register, grammar and lineage

The source presents both a context-free production and a finite-word shift expression. The master keeps their roles explicit:

$$0 \longrightarrow 0, \quad 1 \longrightarrow 1[\phi_+]1[\phi_-]. \quad (12)$$

Brackets preserve the source's branch-state save/restore meaning. A parallel realization can read one parent record and write two candidate slots instead of interpreting a string at runtime. [A, pp. 9–14]

7.1 Literal word variants

For width w with mask $L_w = 2^w - 1$, define

$$B_\oplus(\psi) = ((\psi \ll 1) \wedge L_w) \oplus (\psi \gg 1), \quad (13)$$

$$B_\vee(\psi) = ((\psi \ll 1) \wedge L_w) \vee (\psi \gg 1), \quad (14)$$

$$W_\mu(\psi, J_{jit}) = \begin{cases} 0, & \psi = 0, \\ (B_\mu(\psi) \oplus J_{jit}) \wedge L_w, & \psi \neq 0. \end{cases} \quad (15)$$

A's diagram uses XOR, while another split expression uses OR. They remain two named profiles. The zero guard is the inherited realization of $0 \rightarrow 0$; without it a nonzero XOR input can change zero. When a source variant is selected, both its merge kind and word width must be declared. [A, pp. 10, 13–14][M06]

A finite-word payload may be zero without representing a nonexistent address. The corresponding live flag is separately typed in a lineage profile. The explicit source-word profile above chooses zero as its absorbing payload because that is the inherited word grammar, not because all engine values share that meaning.

7.2 Lineage-address realization

The earlier mathematical repair for genuine child identifiers is retained separately:

$$\ell_b = 2\ell + b = (\ell \ll 1) \vee b, \quad b \in \{0, 1\}, \quad \text{parent}(\ell_b) = \ell_b \gg 1. \quad (16)$$

A right shift is not a right-child constructor. This does not erase the literal shifted-word rule; it assigns a different representation to actual branch records. At depth g , a leading one followed by g branch bits uniquely describes the path. Removing the last bit recovers its parent.

The source calls the structure a binary search tree; the runtime requirement stated here is only a binary *lineage* tree. A search-tree ordering invariant is not supplied. Fixed-width address overflow must be an explicit resource error, not a wrapped lineage or an absorption event.

8 Kinematic evolution, the hinge and gestation

Let z denote a declared numerical state and $f(t, z; \Theta_n)$ its supplied field. The source's RK4 core is retained:

$$k_1 = f(t, z), \quad k_2 = f(t + h/2, z + hk_1/2), \quad (17)$$

$$k_3 = f(t + h/2, z + hk_2/2), \quad k_4 = f(t + h, z + hk_3), \quad (18)$$

$$\mathcal{R}_h(z) = z + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \quad (19)$$

The field, units, step size and any substeps belong to Θ or I_n . The existence of these four stages does not select a physical or domain-specific field. [A, pp. 19–24]

For a planar source realization, the branch hinge adds a signed phase increment:

$$H_b(\rho, \phi) = (\rho, \phi + (-1)^b \delta_\phi). \quad (20)$$

Whether this acts on position phase, orientation, or another kinematic field is a profile choice. The abstract update does not equate a phase offset to a rotated binary representation. Any 16-bit mask-rotation helper remains a separately typed word primitive.

8.1 Which increment feeds OTAN2

A shared sample record names an interval and an increment origin. It may describe a supplied endpoint pair, an accumulated RK4 segment, or a selected branch proposal. The master uses `increment_origin` to distinguish them; it does not add the branch hinge to a derivative increment without declaring that policy. [M07]

In an RK4-linked observation, the output of the integrator and its matching pre-step state give $\Delta\rho$, while the phase lift supplies $\Delta\phi$. The same observation record carries W and the reference-axis frame. This is the direct engine connection of OTAN2 to the numerical state: the readout belongs to the processed sample, not to an unrelated GUI value.

8.2 The V admission layer

The source's womb/gestation terminology names the transition from a proposed state to a retained trace. In the master, an admitted candidate is stored in the next local frontier or word bank; a rejected candidate has a disposition record. Step duration, model event time and a domain's biological or observational delay are not interchanged by their common narrative label.

Boundary sampling and RK4 substeps are also separate choices. A clear sampled predicate certifies its tested sample only. The source supplies no universal continuous-collision contract, so the specification records the chosen samples instead of importing one.

9 Environment, divergence and source-weight profiles

The source writes

$$\nabla \cdot \Psi = e^{-\rho} \left(\frac{\partial \Psi_\rho}{\partial \rho} + \Psi_\rho + \frac{\partial \Psi_\phi}{\partial \phi} \right). \quad (21)$$

This expression is retained as a source equation, with its component interpretation made explicit by the prior formalization: for orthonormal physical radial/tangential components u_r, u_ϕ in a planar log-polar chart,

$$\nabla \cdot u = \frac{1}{r_* e^\rho} (\partial_\rho u_r + u_r + \partial_\phi u_\phi). \quad (22)$$

The source form corresponds to $r_* = 1$. [A, pp. 30–33]

If the numerical state instead supplies coordinate rates $a = \dot{\rho} = u_r/r$ and $b = \dot{\phi} = u_\phi/r$, the same geometric calculation gives

$$\nabla \cdot u = \partial_\rho a + 2a + \partial_\phi b. \quad (23)$$

These are different component conventions, not interchangeable formulas to apply to an untyped bit register. The master keeps `component_basis` in a field descriptor. A quotient topology additionally needs a compatible metric/field definition before this planar formula is applied there.

9.1 A table change is distinct from a field change

An environmental update can change a predicate plane, a numerical field, or a named parameter. Each is a different input. The source's “negative divergence” and Vantablack language does not make a Boolean clear/reset operation a numerical derivative. The local sink is specified by the ASA/disposition equations; field diagnostics remain numerical records.

9.2 Mass/scale and other source labels

A pp. 55–59 also proposes $m_{scaled} = (E/c^2)e^{-\rho}$. The master retains this as a *source-proposed weighting expression*, not as an automatically enabled invariant or a parameter already supplied to RK4. To use a weighting profile, its base value, units, interpretation and position in the data path must be given. No binding from that expression to JK, an external output, or an identity credential is present here.

The same rule governs the source's OMT, logistics and ledger analogies: their labels remain in provenance and the glossary, while a domain adapter must provide its actual state values and equations. The integrated engine's mathematical contracts do not acquire a new input stream simply because a source paragraph names one.

10 JK as the engine's explicit sequential state

A names Operator JK and describes its set and toggle cases. The inherited formal completion is a logical J–K state transition, now a first-class part of the global state rather than an unnamed external host. For every declared JK channel,

$$q^+ = (J_{JK} \wedge \neg q) \vee (\neg K_{JK} \wedge q), \quad q, J_{JK}, K_{JK} \in \{0, 1\}. \quad (24)$$

This is the Boolean model retained from J361, not an inferred voltage-level circuit implementation. [A, pp. 53–55][J361, p. 5]

J_{JK}	K_{JK}	q^+	Mode
0	0	q	Hold
1	0	1	Set
0	1	0	Reset
1	1	$1 - q$	Toggle

The selected tick includes an explicit JK update event. If no event is requested, $Q_{n+1} = Q_n$. If an event is requested, the complete J/K input bank must be supplied and validated, and every channel is evaluated from the *old* Q_n before any new value is committed. This simultaneous-bank rule prevents an iteration order from becoming an accidental dependency. [M08]

10.1 What the latch state controls

The meaning of q is a declared internal mode or retained bit. Its existence does not define an external action. The source does not provide an angular tolerance, hysteresis, or transfer rule connecting OTAN2 to J/K, so no such rule is enabled by this consolidation. J_{jit} , J_{JK} and an alignment discrepancy remain distinct fields.

A future internal policy may explicitly consume a typed observation and produce a J/K pair. Such a policy would be a new named engine profile with its own domain and tests. The master API preserves that extension point, while the default supplied-input rule is already fully defined and usable.

Proposition 10.1 (JK determinism and domain closure). *For every Boolean triple (q, J_{JK}, K_{JK}) , the displayed recurrence returns exactly one Boolean result. Hold is an identity; applying toggle twice recovers q .*

Proof. The four J/K pairs produce the four truth-table rows by Boolean substitution. Each row maps $\{0, 1\}$ to $\{0, 1\}$. For toggle, $1 - (1 - q) = q$; for hold, the result is q directly. $\square$

This is a property of the specified Boolean transition. Missing inputs, incompatible array lengths or failed memory accesses are separate implementation errors, not fifth entries in the truth table.

11 A–W–B–N and the circle-plus blend

C introduces A as the reference eigenvector axis, W as the kinematic interval, B as the “double lowercase phi” descriptor and N as the north polar pinion. These source names are preserved. In the master they are A_{axis} , W , B_ϕ and N_{pinion} . [C, pp. 1–3][P31, pp. 6–7]

The source expression is

$$s_{source} = N \oplus (A + \phi(W)) \pmod{2}. \quad (25)$$

The inherited binary special case requires three *already encoded* bits:

$$s_{blend} = n_b \oplus ((a_b + \beta_b) \bmod 2) = n_b \oplus a_b \oplus \beta_b, \quad n_b, a_b, \beta_b \in \{0, 1\}. \quad (26)$$

β_b here is a binary kinematic encoding, not the OTAN2 angle β . A complete record uses different field names for the two. An axis angle does not become a bit simply by appearing next to the XOR symbol.

n_b	a_b	β_b	s_{blend}
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

11.1 The B-profile remains named, not guessed

C describes B as angular velocity or a transformation matrix, while using the phrase “double lowercase phi.” Neither $\ddot{\phi}$, ϕ^2 nor a specific matrix follows from that phrase. The master keeps a `kinematic_descriptor` and a separately typed `kinematic_result`. The B-profile must define how W is used before its result can be encoded. [C, p. 1][P31, p. 6]

A zero blend result means equality of the encoded parity classes. It means equality of full geometric states only under an encoder that supplies that implication. The master therefore allows a blend record with `unspecified` status when the encoding has not been supplied, while other independently defined engine computations continue.

The source label “precedence 0” is retained as an interpretation/dependency stage: resolve types and encoding before evaluating the bit relation. It is not used as a host privilege level, a global arithmetic precedence override or an elapsed-time value. [P31, p. 12]

12 Three predicates and the hollow-center description

C changes the meaning of its one-bit state as its discussion moves from alignment to sweep limits to a hollow center. The master carries all three forward rather than collapsing them into one control bit. [C, pp. 2–5]

Predicate	Zero means	One means
s_{align}	Alignment/no encoded phase adjustment	Encoded discrepancy/phase-shift condition
s_{limit}	Within the declared sweep envelope	Declared envelope or threshold limit reached
s_{fringe}	In the declared hollow region	On the declared active fringe

Each value has a definition status. No source equation establishes $s_{align} = s_{limit} = s_{fringe}$, and the engine never makes that identification merely because all three values fit a bit. Likewise, an OTAN2 residual is not any one of these bits without an explicit encoding rule. [P31, pp. 8–9]

12.1 Regions rather than invented geometry

Let a supplied region profile classify a hollow region H and active region R_f with $H \cap R_f = \emptyset$. On that classified domain,

$$s_{fringe}(z) = \begin{cases} 0, & z \in H, \\ 1, & z \in R_f. \end{cases} \quad (27)$$

Unclassified points remain unspecified. In particular, “not in the hollow” is not automatically the same as “on the active fringe” when points beyond an outer boundary exist.

C names Cassegrain, Gregorian, ring-focus and phased-array possibilities and asks follow-up questions that are not answered in the source. The master keeps `inner_boundary` and `outer_boundary` as unit-tagged profile fields rather than choosing a reflector or inferring a complete response from a central receiver description. [C, pp. 4–5]

12.2 Connection to the engine aperture

A complete fringe mask M_f may be supplied over the same logical dictionary as ASA. When explicitly enabled, it changes the effective aperture to $M_A \wedge M_f$. That operation is the inherited P31 integration, not a mask silently generated from the three flags. Partial classification cannot masquerade as a total enabled mask. [P31, pp. 9–10]

In default record mode, these source fields are still integrated: they occupy the engine’s committed reference/region record. Missing geometry does not erase a previously valid word result; it is reported at the field that lacks a definition.

13 ASA, NA and the whole-word disposition core

For every logical word, the preserved base contract is

$$a = x \wedge M_A \wedge V_{valid}, \quad (28)$$

$$n = a \wedge M_N, \quad (29)$$

$$h = \text{popcount}(n \wedge M_\partial), \quad (30)$$

$$y = \begin{cases} 0, & h > 0, \\ n, & h = 0, \end{cases} \quad R_{ASA} = (a, n, h, y). \quad (31)$$

This is the documented 3.0/3.1 aperture contract carried into both J361 and I2. ASA remains before NA; the collision quantity is a count, and the disposition is whole-word zeroing. [P31, p. 4][J361, p. 4]

13.1 Logical support and word grouping

The operands are bitsets over the same cell dictionary. x denotes occupancy or a declared selected support; M_A is the ASA aperture; M_N the post-ASA mask; and M_∂ the boundary. V_{valid} suppresses undeclared tail bits. A coordinate key or a full identity digest is not substituted for any mask.

For logical supports X, B , population count of their bitwise AND is $|X \cap B|$. A high-order intersection also counts: the test is $h > 0$, not raw-AND equality to the integer one. For a selected word $n = 255$ and a boundary 128, $h = 1$ and the result is zero, not 127.

13.2 The V sink and its record

Vantablack names this absorbing *state disposition*. The record retains a, n, h and the reason even when $y = 0$. A clear computation that happens to select no bits and an absorbed nonempty word both have final zero but different intermediate records. Consumers must not reconstruct the reason solely from y .

Proposition 13.1 (Fixed-profile projection). *For fixed masks, the final-word map sends zero to zero, never creates an input support bit, and is idempotent under reapplication.*

Proof. Each selection is an intersection with the input. If the first output is absorbed, it is zero and remains zero. Otherwise the output already satisfies the selection masks and contains no boundary intersection, so a second application leaves it unchanged. $\square$

This proposition concerns the final word, not the equality of its first-pass and second-pass histories. The source's negative-divergence or light-absorption descriptions do not change these exact bitset equations.

14 Optional fringe and post-ASA lens profiles

The complete-mask option retains the same four output fields while changing the effective aperture:

$$M'_A = M_A \wedge M_f, \quad R_{fringe} = F_{ASA}(x; M'_A, M_N, M_\partial, V_{valid}). \quad (32)$$

The profile is explicit and the mask names the same dictionary. The neutral setting is $M_f = V_{valid}$. In record mode, no additional mask is applied. [P31, pp. 10–11][J361, p. 12]

Theorem 14.1 (Neutral recovery). *Record mode has the original ASA result. In complete-mask mode, $M_f = V_{valid}$ recovers every field (a, n, h, y) of that result.*

Proof. The first selected word is $x \wedge M_A \wedge V_{valid} \wedge V_{valid} = x \wedge M_A \wedge V_{valid}$. Identical first-stage inputs give identical NA words, boundary counts and final dispositions. Record-only mode leaves the operands unchanged directly. $\square$

14.1 The preserved counterexample

A narrower pre-absorption selection does not guarantee a smaller final output. With $x = M_A = M_N = V_{valid} = 3$ and $M_\partial = 2$, the original record is $(3, 3, 1, 0)$. Selecting $M_f = 1$ gives $(1, 1, 0, 1)$. Removing the boundary bit has released the whole word from absorption. This is a required test of the retained semantics. [P31, p. 11]

14.2 NA after ASA is not an inferred transfer function

A describes liquid lensing and post-ASA numerical aperture. The runtime mask M_N is the inherited Boolean representation. A does not supply a numerical refractive-index table or complete coordinate transfer function. Therefore the master keeps the lens descriptor and any supplied transfer profile separate from the mask operation. [A, pp. 78–81]

A future supplied transform $L_\lambda(z)$ must state its parameters, interpolation, boundary policy, metric/frame effect and exact place in the data path. Applying a transform before selection need not equal applying it after selection. This master does not silently insert such a transform or advertise a lens-offset LUT that is absent from the inspected source contracts.

The distinction matters for OTAN2 as well. If a transform changes a path or tangent frame, the orientation input must identify whether it was taken before or after that transform. Reusing the same field name is not sufficient to make those samples equivalent.

15 OTAN2 at the center of the integrated engine

O first discusses ordinary atan2 and phase projection, then records the author's spelling correction and supplies the literal operator. The master retains that final custom name and distinguishes it from the preparatory alternatives. [O, pp. 1–6]

Source default / source_ratio

$$\text{otan2}(\Delta\phi, \Delta\rho) = \arctan\left(\frac{\Delta\phi}{\Delta\rho}\right), \quad \Delta\rho \neq 0. \quad (33)$$

For the source's moving reference-axis angle α ,

$$e_{\text{source}} = \text{otan2}(\Delta\phi, \Delta\rho) - \alpha. \quad (34)$$

The raw subtraction is retained; it is not silently wrapped or converted to a JK input.

$\Delta\phi$ is the angular change in radians and $\Delta\rho$ a log-radius change over the corresponding interval. The source's $\Delta\theta$ is an alias for the selected lower-case phase coordinate. The input order is angular increment first, radial increment second. [O, pp. 6–7]

15.1 The explicit directed completion

The inherited opt-in profile `directed_completion_1` uses

$$\beta_{\text{dir}} = \text{wrap}_{2\pi}(\text{atan2}(\Delta\phi, \Delta\rho)), \quad (\Delta\rho, \Delta\phi) \neq (0, 0). \quad (35)$$

This is a J361 completion, not a replacement definition of the source operator. The literal ratio forgets simultaneous sign reversal; the directed profile distinguishes it by a half-turn. A label such as “orthogonal” in O does not add a hidden $\pi/2$ rotation. [J361, p. 8]

15.2 First-class engine observation

Every requested sample has one OTAN2Record, even when its value is undefined. It identifies the sample, epoch, increment origin, selected profile, units/frame, interval, axis, raw orientation and residual views. Its invariant records reference the same sample identifier. Consequently, OTAN2 is integrated into the engine state and commit, not delivered as a separate accessory.

Calling it central does not invent a closed-loop control law. The numerical readout is computed from the actual supplied/integrated sample; ASA and JK remain governed by their displayed equations and explicit inputs.

16 Phase lifts, moving axes and the angular domain

Define the principal representative for period $T > 0$ by

$$\text{wrap}_T(u) = u - T \left\lfloor \frac{u + T/2}{T} \right\rfloor \in [-T/2, T/2). \quad (36)$$

The engine retains three views when the corresponding values are defined:

$$e_{\text{raw}} = \beta - \alpha, \quad e_{\text{directed}} = \text{wrap}_{2\pi}(e_{\text{raw}}), \quad e_{\text{line}} = \text{wrap}_{\pi}(e_{\text{raw}}). \quad (37)$$

The positive half-period endpoint is represented negatively. A principal projection is not a reconstruction of lost winding. [J361, pp. 8–10][I2, contract]

16.1 Explicit angular increment policies

If phases are recorded modulo 2π , the actual change has the form

$$\Delta\phi = \phi_1 - \phi_0 + 2\pi k, \quad k \in \mathbb{Z}. \quad (38)$$

The two wrapped samples do not determine k . The retained policies are an explicit increment, known turn difference, or a deliberately selected principal-difference convention. For $359^\circ \rightarrow 1^\circ$, direct subtraction is -358° , while principal difference is 2° . The record must state which was selected. [J361, p. 9]

16.2 Static chart; moving pinion; moving reference

The pinion's path supplies the increment vector in the static chart. The reference supplies α in the same log-tangent frame and scaling convention. Changing only α by η gives

$$e'_{\text{raw}} = e_{\text{raw}} - \eta. \quad (39)$$

This is covariance, not an invariant. The stored chart and aperture do not move merely because a residual changes. [O, pp. 6–7]

The term “eigenvalue axis” is retained as source nomenclature for a preferred reference orientation. No matrix or eigenpair is supplied by O. A provided axis in another frame stays `frame_unspecified` until the corresponding conversion is declared; an eigenvector is not calculated from an absent matrix.

16.3 Singularities and numeric range

The literal ratio is undefined at $\Delta\rho = 0$. Both profiles report a zero-increment status when both components vanish. The implemented I2 component also rejects nonfinite operands, ratio overflow and nonzero-ratio underflow to zero. These statuses remain metadata, not zero-valued angles. The formal real-valued definition and a finite implementation's range are recorded separately.

17 The invariant bank: R, W, P and S

The v3.6 invariant controls are a bank of named transformation tests over the same OTAN2 sample. They distinguish an equality proved under assumptions from the finite arithmetic comparison of a particular input. Four cards are preserved-value identities; two further cards are profile-dependent laws. [I2, pp. 7–11]

17.1 R — common radial scale

Scaling both radii about the same origin by $c > 0$ adds $\ln c$ to both log radii:

$$(\rho_1 + \ln c) - (\rho_0 + \ln c) = \Delta\rho. \quad (40)$$

With $\Delta\phi$ and α unchanged, either defined orientation profile and its relative readout are unchanged. The same origin, common scale and common reference length are preconditions. This is the explicit algebra behind O's scale statement. [O, pp. 3, 6]

17.2 W — common interval

For a finite positive common interval W ,

$$\frac{\Delta\phi/W}{\Delta\rho/W} = \frac{\Delta\phi}{\Delta\rho}. \quad (41)$$

Dividing both components by that interval preserves orientation. It does not license comparing components drawn from different intervals without conversion. The clock interval is not inferred from an author identifier.

17.3 P — phase origin

For consistently lifted phases and a common shift γ ,

$$(\phi_1 + \gamma) - (\phi_0 + \gamma) = \Delta\phi. \quad (42)$$

The same lift must be used for both endpoints. Independently rewrapping each endpoint can introduce a branch-cut change in a naive subtraction. That is why P names a lift precondition rather than hiding it in a modulo operation.

17.4 S — step scale / spiral pitch

For $s > 0$, the pair $(s\Delta\rho, s\Delta\phi)$ has the same direction and source ratio. For the illustrative chart relation $\phi = k\rho + c$, every nonzero radial segment has $\Delta\phi/\Delta\rho = k$ and the literal value $\arctan k$. This derives the constant-pitch statement without adding an image correspondence or a field estimator. [O, p. 6]

These identities do not assert that all coordinates or all engine outputs stay fixed under the same transformation. The declared object preserved is the appropriate orientation/residual, with the rest of its preconditions held constant. Positive step-scale invariance is not the same operation as changing both signs, which belongs to V.

18 The invariant bank: F, V and panel semantics

18.1 F — joint frame rotation

Write $v = (\Delta\rho, \Delta\phi)^T$ in the common tangent frame. Rotate v by η and advance α by the same angle. The directed relative class is unchanged modulo 2π . The literal ratio preserves the corresponding *line class* modulo π , provided both literal denominators are nonzero.

$$\text{wrap}_{2\pi}(\beta'_{dir} - (\alpha + \eta)) = \text{wrap}_{2\pi}(\beta_{dir} - \alpha), \quad (43)$$

$$\text{wrap}_{\pi}(\beta'_{source} - (\alpha + \eta)) = \text{wrap}_{\pi}(\beta_{source} - \alpha). \quad (44)$$

Raw literal residual equality does not hold across every branch choice. The source example $v = (1, 0)$, $\alpha = 0$, rotated by $3\pi/4$ gives raw literal residual $-\pi$ but the same line-class residual zero. [I2, F-card contract]

18.2 V — direction reversal

Negating both increments preserves their quotient, so the literal orientation is unchanged wherever defined. The directed completion instead changes by a half-turn:

$$\text{otan2}_{source}(-\Delta\phi, -\Delta\rho) = \text{otan2}_{source}(\Delta\phi, \Delta\rho), \quad (45)$$

$$\text{wrap}_{2\pi}(\beta_{dir}(-v) - \beta_{dir}(v)) = -\pi. \quad (46)$$

The negative representative is the selected half-open convention. V is therefore a *profile law*, not a claim that reversal has the same interpretation in both profiles. [I2, V-card contract]

18.3 PASS, FAIL and UNDEFINED

The I2 comparison tolerance is 10^{-10} radians. A finite comparison within that tolerance reports PASS; one outside reports FAIL; a failed domain/precondition reports UNDEFINED with its reason. Undefined is never counted as a successful invariant check.

The inherited F-probe guards near-vertical source vectors by

$$|\Delta\rho'| \leq 64 \epsilon_{machine} \text{hypot}(\Delta\rho, \Delta\phi). \quad (47)$$

This is a numerical probe guard, not a redefinition of the source operator's exact nonzero-denominator domain. The R/P probe baselines 0.75 and 0.25, and the default transformation magnitudes, are test constructors from I2 rather than source constants.

The complete six-card result is committed with the sample it checks. A FAIL is a diagnostic record; it does not silently change masks, JK inputs or future time steps. Any different response policy must be explicitly selected and versioned.

19 The total integrated tick

The master transition uses one immutable input/configuration snapshot and one commit. The following is a **newly declared assembly of the retained operators**, not an order claimed to have appeared verbatim in a source transcript. The preserved local orders remain unchanged inside it. [M03, 09]

Stage	Work	Dependency / output
P0	Resolve the epoch snapshot	Verify identity/profile references, dictionary IDs, types, required inputs and budgets. Do not alter live state.
P1	Advance enabled state lanes	Apply the selected raw-word rule or lineage/hinge/RK4 profile; optionally advance a programmed machine prefix. Disabled lanes retain their old state.
P2	Bind the logical word input	Select an explicit input bank, retained bank or a declared emitter output. No automatic conversion from a path, digest or sensor label to occupancy is assumed.
P3	Resolve the sample record	Bind interval, increment origin, phase policy, common frame, A–W–B–N values and predicate statuses.
P4	Evaluate OTAN2 and invariants	Compute the literal or selected completed value and its residual views. Compute the six diagnostics from that same sample.
P5	Apply the aperture contract	ASA, optional complete fringe, NA, POPCNT and whole-word disposition, producing (a, n, h, y) per logical word.
P6	Evaluate the JK bank	Use explicit J/K inputs and the old Q_n on a declared event; otherwise retain Q_n . No undeclared angular feedback.
P7	Assemble and commit	Publish next state, outputs, source/status records, manifests and one ledger extension in canonical order.

The source-local word precedence remains shift, merge/jitter, hinge where applicable, followed by its named boundary stage. The blend's precedence-zero label remains type/encoding resolution. ASA remains before NA. OTAN2 is evaluated after its required increment/frame are resolved, and its records are committed with the epoch.

19.1 Independent lanes, explicit links

The core can execute word filtering without a geometric lane, or a geometric lane can supply a sample through a declared emitter. A callable field or emitter is a configuration input, not a hidden implementation gap. The master does not force every operation into a serial dependency when its source defines no such dependency.

An identity carrier for the word bank and explicit observation increments make a complete non-geometric tick fully specified. Section 29 executes that integrated case with JK state, blend, OTAN2, all six invariants, aperture output and a ledger record.

20 Carrier binding: geometry into the word kernel

A total assembly needs an explicit link between geometric candidates and logical words, rather than a guessed conversion. M1 therefore defines the optional `lineage_words` carrier below as a **new master adapter**. Its use must be selected in the configuration; direct/retained word-bank modes remain separate. [M15]

Let the valid candidate registry be $\widehat{F}$. A declared chart quantizer maps each candidate C to a cell with logical word $w(C)$ and bit $b(C) \in \{0, \dots, 31\}$. The emitted occupancy is

$$\text{emit}_{\Theta}(\widehat{F})[q] = \bigvee_{C \in \widehat{F}: w(C)=q} 2^{b(C)}. \quad (48)$$

Candidates outside the selected chart are handled by the geometric profile's explicit boundary disposition before emission. The adapter is not defined on a candidate with missing coordinates, unresolved chart identity or an invalid lookup result.

Apply the unchanged ASA/NA/fringe/whole-word map to that bank, producing Y . Admission back into the lineage registry is

$$F_{n+1} = \text{admit}(\widehat{F}, Y) = \{C \in \widehat{F} : (Y[w(C)] \gg b(C)) \wedge 1 = 1\}. \quad (49)$$

The full registry retains every candidate identity and lineage key. If two candidates share one cell, one occupancy bit represents that support, while two registry records remain distinct. This adapter shares the cell/group disposition between them; it does not invent an individual classification from a single bit.

Proposition 20.1 (Word–registry consistency). *In the selected carrier, with $X = \text{emit}(\widehat{F})$ and final ASA output Y , re-emitting the admitted registry gives $\text{emit}(F_{n+1}) = Y$.*

Proof. ASA selects only bits already present in X , so every one in Y has at least one candidate at that cell. Admission keeps precisely the candidates whose selected cell bit is one in Y . Their union contains every such bit and no other bit. Multiple candidates at one cell do not change that union. $\square$

OTAN2 samples can refer to these same candidate IDs, but their increments are resolved from the declared unquantized state segment before this support projection unless a separate quantized-observation profile is explicitly chosen. Absorption does not delete the corresponding diagnostic sample history.

This adapter supplies the missing *declared* link for an integrated geometric-to-word tick. It is not attributed to an undocumented source equation and does not add an external sensor, identity matcher or actuator interface.

21 Transition equations and commit semantics

Let $\widehat{F}$, $\widehat{U}$ be the validated proposed states of enabled evolution lanes, or their old states when disabled. In the selected `lineage_words` carrier, the admitted frontier is defined in Section 20; otherwise the independently selected frontier profile supplies its next state. Let $\widehat{X}$ be the selected logical input bank. The chosen carrier/emitter must already use the mask dictionary. Define

$$R_{A,n} = \{F_{ASA}(\widehat{X}[q]; \Theta_n[q])\}_{q=0}^{L_w-1}, \quad (50)$$

$$X_{n+1}[q] = R_{A,n}[q].y, \quad (51)$$

$$Q_{n+1}[j] = \begin{cases} \text{JK}(Q_n[j], J_n[j], K_n[j]), & \text{JK event,} \\ Q_n[j], & \text{otherwise,} \end{cases} \quad (52)$$

$$D_{n+1} = \langle P_{blend,n}, P_{region,n}, P_{OTAN2,n}, \text{Inv}_n \rangle. \quad (53)$$

The observation and invariant fields are fully part of S_{n+1} , even though their default effect on the aperture and JK recurrence is zero. [M09]

Construct an event body E_n with the old epoch, interval, dictionaries, selected profiles, core results, old/new JK values, enabled-lane outcomes and D_{n+1} . The ledger rule produces L_{n+1} . Then

$$S_{n+1} = \langle G, n+1, t_n + W_n, \Theta_n, F_{n+1}, X_{n+1}, Q_{n+1}, \widehat{U}, L_{n+1}, D_{n+1} \rangle. \quad (54)$$

Any accepted configuration change becomes a new versioned snapshot; it is never written into tables that an unfinished kernel still reads.

21.1 Fatal inputs versus undefined observations

Invalid required word types, missing enabled-mask definitions, incompatible dictionaries, failed required allocations or nonfinite active numerical evolution reject the epoch before commit. The persistent state and ledger head stay unchanged. An out-of-domain OTAN2 sample, by contrast, produces its defined error/status record; it need not invalidate an otherwise valid independent ASA result.

An absent B encoder leaves the blend unspecified. An absent moving axis can leave the orientation defined while the relative residual remains unspecified. These are legitimate committed records describing available and unavailable information, not zero-filled substitutes.

21.2 Why a commit is necessary

Updating the old JK bank in place while other channels read it, publishing half of a new aperture table, or logging a state before its required work completes would introduce an order-dependent engine. M1 excludes those cases through snapshot reads and staged writes. This is a logical transaction contract; a concrete storage backend must implement equivalent visibility and failure behavior.

22 Engine records and interface schema

The complete record has a stable core and separately typed groups. This is a logical schema; fixed offsets for an unseen historical binary are not asserted.

Record group	Required content / interpretation
version	Engine family v3.6, master revision M1, individual component/profile identifiers.
epoch	Epoch index, model interval, model time, input snapshot and parent ledger head.
identity dictionary	Full genesis digest and registry references; source attribution stored separately. Chart profile, dimensions, quantization, tail masks, word grouping, topology and storage permutation.
evolution	Enabled lanes, incoming and proposed states, field/profile IDs, sample paths and resource outcomes.
core	Canonically ordered (a, n, h, y) words; fringe mode and mask identity where enabled.
jk	Old/new q , explicit J/K operands, clock event and the mode's declared role.
pinion	A-W-B-N references, kinematic/encoding profile, three predicate values and their statuses.
otan2	Profile, sample ID, increments, phase policy, frame, axis, orientation status and residual views.
invariants	R/W/P/S/F/V IDs, assumptions, transform values, comparison class, expected/observed/error and status.
provenance	Source references, configuration digest, code/numerics profile and event serialization version.

22.1 Stable names for known status aliases

I1 used `undefined_ratio` in its document table; the supplied I2 code returns `ratio_undefined`. M1 uses the executable I2 name as the canonical field value and records the former as an import alias. A similar rule maps narrative “increment unspecified” to the precise missing-policy or missing-turn status. This is an explicit schema reconciliation, not a silent change to the source formula. [M10]

Zero increments, missing axis, missing frame, missing winding policy, numerical range failure and nonfinite inputs remain separate. Mask-mode incompleteness is a required-input error, while a record-only region may be unspecified. The schema's status and profile fields prevent consumers from confusing those cases.

The four ASA integers may retain a 16-byte low-level layout. The master record containing them need not be exactly 16 bytes: typed provenance and numerical observation data are additional information, not quantities compressed into the same word.

23 Packed cells, valid bits and Morton textures

Let $W_p^{logic} = \lceil P/32 \rceil$ be the number of angular words per radial row. To avoid confusing this count with the time interval W , implementation records call it `words_per_row`. Word (i, w) contains cells $(i, 32w + b)$ in its low-to-high bits $b = 0, \dots, 31$.

Pad row and word counts upward to multiples of eight, giving R_p and C_p . The last logical word carries

$$V_w = \begin{cases} 2^{P \bmod 32} - 1, & w = W_p^{logic} - 1 \text{ and } P \bmod 32 \neq 0, \\ 2^{32} - 1, & \text{otherwise.} \end{cases} \quad (55)$$

All padded texels are zero. No implementation shifts a 32-bit value by 32 to construct the full mask; the full-word case is explicit.

23.1 Application-visible Morton map

Within an 8×8 word tile, radial bits occupy even address positions and angular-word bits odd positions:

$$z(i_0, w_0) = \sum_{k=0}^2 \left(\text{bit}_k(i_0)2^{2k} + \text{bit}_k(w_0)2^{2k+1} \right). \quad (56)$$

With $i_0 = i \bmod 8$ and $w_0 = w \bmod 8$,

$$\pi(i, w) = 64 \left(\lfloor i/8 \rfloor (C_p/8) + \lfloor w/8 \rfloor \right) + z(i_0, w_0). \quad (57)$$

This retains A 's bit-order convention and the prior word-tile realization. Logical output remains row-major $q = iW_p^{logic} + w$. [A, pp. 85–87][P31, p. 4]

Theorem 23.1 (Swizzle equivalence). *If every stored input plane satisfies $M_\pi[\pi(i, w)] = M[i, w]$ and output groups remain canonical, changing between a bijective linear and Morton layout changes no ASA result.*

Proof. The mapping relation returns the same four or five logical input words at every logical location. The evaluator is the same function on those words, and the output address is independent of the storage permutation. Equality holds word by word. The within-tile inverse extracts even and odd bits, proving the local bijection; row-major tile indexing completes it globally. $\square$

A swizzle moves data, not the underlying meaning of a bit. It does not change the phase frame, reorder JK events or turn a native program instruction into texture data.

24 GPU backend and the laptop memory profile

The retained target is the **RTX 5070 Ti Laptop GPU with 12 GB GDDR7**. NVIDIA's compute-capability listing assigns the RTX 5070 Ti family to 12.0, and CUDA 12.8 introduced compiler support for SM_120. The implementation target should be checked against the actual selected device rather than inferred from the document title. [H1, 2, 3]

The source's texture request is mapped to read-only data resources. A linear unsigned-word texture uses an explicitly supplied address and `tex1Dfetch`; a global-read backend evaluates the same expression from the same words. CUDA's linear-resource texture addressing and filtering modes do not replace application bounds checks. [H4]

The integrated allocation categories are

$$M_{total} = M_{word\ kernel} + M_{fringe} + M_{frontier} + M_{JK} + M_{OTAN2} + M_U + M_{staging} + M_{margin}. \quad (58)$$

For stored plane count $S = R_p C_p$ and logical output count $L_w = R \lceil P/32 \rceil$, the *base* four-input-plane/16-byte-output word kernel occupies

$$M_{word\ kernel} = 16S + 16L_w. \quad (59)$$

At the 128×1024 review profile this is 131,072 bytes. That is not the whole engine's memory use. Enabled fringe and numerical/frontier/record storage must be added before admission. [M11]

The earlier review defaults retain a 64 MiB planning margin, a 512 MiB user ceiling and a 1,536 MiB total-device reserve. With probed free bytes F and total bytes T , require

$$T > M_{reserve}, \quad M_{total} \leq \min(B_{user}, 7 \lfloor F/10 \rfloor, T - M_{reserve}). \quad (60)$$

An actual allocation failure remains an explicit error even after admission, because available resources can change.

24.1 The ring-0 boundary

The master adopts ordinary user-mode CUDA through the installed NVIDIA driver. CPU kernel-mode privilege and a CUDA compute kernel are different interfaces. No new host ring-0 module, driver patch or code-in-texture mechanism is specified. The source's "0-ring cache" remains an architectural label, not a recovered host ABI. [A, pp. 85–87][H5]

Input snapshots remain immutable until dependent work completes. Mutable outputs, JK banks and tapes use separate storage. Cache residency and speedup are measurement outcomes, not engine invariants. This PDF specifies the backend contract; it does not report a newly compiled or benchmarked full GPU engine.

25 Faceted input, mosaic records and timing

B supplies the ommatidia/facet organization, edge reduction and central mosaic-stitcher narrative. A's final pages attach it to the larger atomOS context. The master preserves that organization as producer ports feeding a common declared dictionary. [B, pp. 7–8][A, pp. 87–91]

For local synthetic or otherwise supplied producer words X_i , a Boolean union adapter may declare

$$X = \bigvee_i X_i. \quad (61)$$

OR is associative, commutative and idempotent, so the abstract set result does not depend on enumeration or duplicate contributions. This does not supply cross-producer geometry, calibration or timing: those must already agree with the chosen dictionary.

If unknown differs from known-clear, known-cell masks K_i are needed as separate metadata or planes. A selected adapter can retain $K = \bigvee_i K_i$ and $X = \bigvee_i (X_i \wedge K_i)$. Missing input must not silently become an observed zero. This is an inherited interface discipline, not a new live sensor implementation.

25.1 The source hardware port

The IWR1642 and optic-lobe/HWA/DSP names belong to B's proposed upstream organization. The source supplies no complete firmware, calibration, frame schema or deployed acquisition artifact. The master therefore names the input port and its required output type without adding register configuration or external data acquisition. The core and OTAN2 can be exercised from local synthetic records independently of that port.

25.2 250 Hz and 128 Hz remain distinct

B's 250 Hz and A's 128-tick labels are separate source targets. Their exact periods are 1/250 s and 1/128 s. The shared repetition interval is

$$T_c = \frac{1}{\gcd(250, 128)} = \frac{1}{2} \text{ s}, \quad (62)$$

containing 125 events of the first stream and 64 of the second. An exact base quantum 1/32000 s represents the periods as 128 and 250 quanta. This is arithmetic about the named rates, not a newly configured hardware clock. [B, p. 7][A, p. 83]

Any multi-rate adapter must state whether it batches, accumulates or otherwise maps events. OTAN2's W card requires a common interval for the two compared increments; it does not solve a missing multi-rate association by cancellation.

26 Provenance, replay and the committed ledger

The retained ledger idea binds state records to the genesis and prior history. The master event body includes profile IDs, input/table digests, ordered core/JK results and the integrated OTAN2/invariant records. Denote its declared canonical encoding by C and a master-specific domain separator by d_{36} :

$$\ell_{n+1} = \text{SHA256}(d_{36} \parallel \ell_n \parallel \text{len64}_{BE}(C(E_n)) \parallel C(E_n)). \quad (63)$$

The exact byte format belongs to a serializer profile. An existing profile may be retained unchanged for its component records, while the master envelope receives a distinct version. This avoids claiming that a new envelope has the same byte hash as an older record. [M12]

26.1 Two useful reproducibility notions

Semantic replay compares the same logical state and result values under the same declared numerical rules. **Byte replay** also requires identical canonical encoding and numerical representation. A timing measurement can change between otherwise equivalent runs; decide explicitly whether such measurements belong in a sealed event or a separate benchmark record.

The I2 component uses floating-point arithmetic and transcendental functions. Integer mask/JK results can be compared exactly; cross-runtime floating comparisons use declared tolerances. A digest equality claim cannot erase a numeric-representation difference, and a tolerance comparison is not the same as byte identity.

26.2 What absorption retains

A Vantablack output word may be zero while its nonzero selected support and hit count remain in the event. A numerical or input error may reject a tick without extending the authoritative state ledger, while an out-of-band diagnostic records the rejection. The master separates those two channels so that a failed transaction does not look like a successful state update.

Proposition 26.1 (Conditional deterministic replay). *For identical state, input, configuration, enabled policies, evaluation semantics and canonical record ordering, the master tick yields identical proposed results and the same ledger successor.*

Proof. Each enabled component is a deterministic function of its declared input under the stated semantics. Snapshot reads remove scheduling-dependent mutation. Canonical ordering fixes event construction, and the digest is a deterministic function of the resulting bytes and prior head. $\square$

A retained trusted head supplies an integrity reference; an unanchored self-consistent chain is not itself authentication. Full input truth, signing and trusted timestamps are separate contracts. This keeps the source's provenance vocabulary without asserting an unprovided authority mechanism.

27 The retained programmable-kernel extension

Earlier project formalizations introduced a programmable extension to give “universal kernel” a precise computational meaning. That extension is retained here as an optional engine lane U, clearly distinct from a claim that fixed shift/XOR or OTAN2 alone is universal. It is restated as a mathematical construction rather than credited to an equation absent from the source transcripts. [M13]

A deterministic machine has a finite control set Q_U , a finite alphabet $\Gamma = \{0, \dots, K-1\}$ with blank 0, halt states H , and a transition table

$$\delta : (Q_U \setminus H) \times \Gamma \rightarrow Q_U \times \Gamma \times \{-1, 0, 1\}. \quad (64)$$

Its configuration is (q_U, p, T) with head $p \in \mathbb{Z}$ and tape $T : \mathbb{Z} \rightarrow \Gamma$. For a supplied rule $\delta(q_U, T(p)) = (q'_U, b, d)$,

$$(q_U, p, T) \mapsto (q'_U, p + d, T[p \mapsto b]). \quad (65)$$

q_U is a program-control state and must not be confused with the separate JK bank Q_n .

27.1 Packing does not remove logical addresses

With $k = \lceil \log_2 K \rceil$, encode tape symbol $T(i)$ in the k bits starting at logical address ki . A sparse word dictionary represents absent words as blank zero. Negative addresses require mathematical floor/remainder semantics; finite device storage uses an explicit origin offset and range checks.

The instruction table can be read-only data while the simulated tape is writable. A table entry is not native code placed in a texture cache: an interpreter performs the corresponding read/write/move operations. Multiple independent tapes can execute in separate lanes; one dependent machine sequence remains ordered.

27.2 Master scheduling

When U is enabled, the epoch supplies a finite transition budget. Completed transitions update a proposed U state; reaching the budget returns a running prefix status, not a halt. A missing rule or unavailable required tape range is an explicit stop/error condition under the selected U profile. No partial individual transition is committed. A required lane error rejects the master epoch; a declared finite budget boundary does not invent nontermination.

The master can keep U disabled without removing the definition from the total architecture. Enabling it does not change ASA, JK or OTAN2 unless an additional explicitly declared local program interface connects their records.

28 Computational simulation and its stated scope

Theorem 28.1 (Machine simulation in the U profile). *Assume logical integer addresses and enough storage for each finite execution prefix. Every deterministic machine with finite control and finite alphabet as defined in Section 27 can be encoded so that the U interpreter preserves its tape, head, control state and halting behavior after each defined transition.*

Proof. Number the finite symbols and states. Store the machine's transition table unchanged under those codes. Pack each initial symbol into its k -bit tape block, and copy its initial head and control state.

Use the invariant that every decoded block equals the corresponding machine symbol and that the head and control state agree. It holds initially. If it holds before a step, the interpreter reads the same symbol, selects the same transition, overwrites exactly the current symbol block, and applies the same head and control updates. Distinct fixed-size blocks are disjoint, so other symbols are unchanged. The invariant holds after the step. Induction gives every finite prefix and hence the same entry into a halt state. $\square$

This is the retained meaning of programmable computational universality. It relies on addressable state and a supplied transition table. A finite wrapped polar dictionary cannot replace an unbounded logical tape without eventually aliasing distinct logical cells.

28.1 Finite resources and the master

A device or host execution uses finite memory and a finite budget. It agrees with the abstract machine only over the prefix for which those resources and transition definitions suffice. A resource stop is therefore carried as a status in the engine's provenance, not mislabeled as an abstract halt or as completion of an arbitrary computation.

The universality proof is not a proof about an unavailable compiled v3.6 runtime. It is a property of the explicit mathematical interpreter. The current PDF consolidates that construction with the rest of atomOS so that its assumptions and state boundaries are visible in one place.

28.2 Relation to the invariant controls

The OTAN2 invariant bank tests selected transformations of orientation samples. It neither makes every program invariant nor proves arbitrary program termination. A future U program may inspect engine records under an explicit input adapter, but its own behavior still requires its own specification. Preserving this distinction allows the engine to have both a general interpreter lane and a small, precisely defined bank of invariants without conflating their roles.

29 A complete worked engine tick

This worked configuration enables the logical word bank, supplied-input JK updates, binary blend, OTAN2, all six invariant checks and the ledger. The geometric and programmable lanes are explicitly disabled for this tick, so they retain their state. The carrier is the identity input-bank profile. These are test choices, not constants in the source transcripts. [M14]

29.1 Initial state and inputs

Choose epoch 0, model time 0 and interval $W = 0.125$. The four initial logical words are $X_0 = (15, 3, 0, 255)$; the JK bank is $Q_0 = (0, 1, 0, 1)$. On the declared JK event, supply

$$J = (0, 0, 1, 1), \quad K = (0, 1, 0, 1). \quad (66)$$

The blend encoding is $(n_b, a_b, \beta_b) = (1, 0, 1)$, so $s_{blend} = 0$. Alignment interpretation is explicitly adopted for that encoded value; no limit or fringe predicate is inferred from it.

Supply the OTAN2 sample

$$\Delta\rho = 0.5, \quad \Delta\phi = 0.5 \text{ rad}, \quad \alpha = \pi/12, \quad (67)$$

in the `log_tangent` frame under `explicit_increment` and `source_ratio`. Its output is

$$\beta = \pi/4 = 45^\circ, \quad e_{raw} = \pi/6 = 30^\circ. \quad (68)$$

For this input both principal residual views also equal 30° .

Word	x	M_A	M_N	M_∂	V_{valid}	M_f
0	15	12	10	0	15	15
1	3	3	3	2	3	3
2	0	15	15	0	15	15
3	255	255	255	128	255	255

The fringe mask is neutral in every row. The resulting core records are

$$R_A = [(12, 8, 0, 8), (3, 3, 1, 0), (0, 0, 0, 0), (255, 255, 1, 0)]. \quad (69)$$

The final word bank is $X_1 = (8, 0, 0, 0)$, and the JK truth table gives $Q_1 = (0, 0, 1, 0)$. Word 2 is empty; words 1 and 3 are absorbed. Their equal final zero does not erase that distinction.

30 The same tick's controls, commit and variants

30.1 Invariant results belong to the tick

Using I2's default probe parameters, all six checks pass for the preceding sample. R compares a common ln 10 offset, W compares the common positive interval, P compares a common phase shift, S scales both components by 3, F rotates both vector and axis by 0.4 rad, and V checks reversal under the literal profile. Each record names the comparison class; F's is the line class, not an unrestricted raw-residual equality.

After validation, the commit publishes epoch 1, time 0.125, the new word and JK banks, the encoded blend, the OTAN2 observation and its six diagnostics, and a new ledger head. The old state is unchanged. Re-evaluating the same old state and inputs with the same encoding yields the same proposed record and successor head.

30.2 An explicit fringe variant

Change only $M_f[0]$ from 15 to 7 and $M_f[1]$ from 3 to 1. The first two records become

$$(4, 0, 0, 0), \quad (1, 1, 0, 1), \quad (70)$$

so the next word bank is (0, 1, 0, 0). The JK output remains (0, 0, 1, 0) because its explicit inputs have not changed. OTAN2 and the invariant sample also remain unchanged. The event and ledger head change because the selected mask profile and core results changed.

30.3 An undefined source-ratio variant

Change the observation only to $\Delta\rho = 0$, $\Delta\phi = 1$. The source profile returns `ratio_undefined`, its angle/residual values are null, and the bank checks are UNDEFINED with the corresponding reason. The valid ASA and JK inputs still yield their original core/state result. A numerical zero angle is not substituted for the missing value.

30.4 A rejected transaction variant

Changing an input word to -1, or supplying a nonpositive model interval, violates a required contract. The master does not publish partial next state, advance epoch/time or extend its authoritative ledger. This is different from a successful tick that records an undefined observation.

These variants were exercised in the new master-assembly audit. They demonstrate the integrated state ownership, not a graphical-panel-only operation and not an assertion that optional geometric or hardware paths were executed.

31 Compatibility and engine-wide invariants

The integration preserves well-defined component behavior while changing the total record structure. The compatibility target is stated per projection, not by claiming every old byte layout remains identical.

Theorem 31.1 (Core and JK noninterference). *Fix the selected logical input bank, masks, topology/grouping and explicit JK event/inputs. Changing only an OTAN2 sample, a reference-only B/region record or invariant probe parameters leaves the ASA output bank and JK next-state bank unchanged under the default no-feedback profile.*

Proof. The ASA expressions use only the fixed word operands and masks. The JK recurrence uses only the fixed old state, event and J/K inputs. Neither reads the changed fields in this profile. Their projected outputs are therefore identical. The observation records and ledger may differ and are not part of the claimed invariant. $\square$

Invariant / recovery rule	Exact scope
Genesis preservation	Successful ordinary ticks retain G; a new genesis is a separately named initialization event.
Neutral-fringe recovery	$M_f = V_{valid}$ gives the base four-word result.
Storage-layout equivalence	Linear/Morton reads recover identical logical words and canonical outputs.
One-bit domain closure	Defined blend/JK outputs are bits; definition status remains metadata.
No invented support	Final ASA support is inside the <i>selected input bank</i> ; a prior evolution lane may itself create proposals under its own rule.
No live-source resurrection	The selected source-word/lineage guard is honored; this is not a prohibition on a separate JK bit being set.
Transaction isolation	Fatal required-input errors publish no partial next epoch.
Observer identity	Each OTAN2/invariant record refers to its declared sample, frame, interval and profile.

Whole-engine evolution is *not* idempotent merely because a fixed-mask ASA projection is. Time and epoch advance; a JK toggle changes state; a numerical field changes coordinates; a ledger gains a record. Likewise, the source ratio's reversal property is not a statement that every other engine field is invariant under reversal.

This separation is essential for the requested small control panel of invariants: each card states what quantity should remain fixed, under which transformation, and which remaining parts of the engine may legitimately change.

32 Error taxonomy and required profile inputs

Condition	Record / status	Master disposition
Both increments zero	zero_increment	Commit unavailable orientation if other required lanes are valid.
Literal zero denominator	ratio_undefined	Preserve missing angle; do not silently select directed completion.
Missing reference axis	axis_unspecified	Raw orientation can remain available.
Unmatched coordinate frame	frame_unspecified	Do not subtract incompatible angles.
No selected turn policy	turn_information_unspecified	Do not guess the accumulated phase.
Nonfinite / unrepresentable observation	nonfinite_input / numerical_range	Preserve the reason as observation status.
Incomplete optional mask while enabled	Required-mask error	Reject epoch; record-only mode is a separate valid choice.
Invalid word, dictionary or JK bank	Type/shape error	Reject epoch before commit.
Geometric lane lacks field or exceeds budget	Required-profile/resource error	No fabricated trajectory or partial committed generation.
Programmable step budget reached	Running finite prefix	Do not relabel as halt; commit only the defined completed prefix when enabled.
GPU allocation / launch failure	Backend error	No success record and no covert CPU substitution.
Finite invariant comparison outside tolerance	FAIL diagnostic	Commit its measured result; no undeclared control response.

32.1 Parameters the sources do not determine

A complete numerical instance supplies any enabled field, axis/frame conversion, B encoder, fringe geometry, NA transfer profile, emitter/cell mapping, JK input policy and serializer details. C leaves waveform and receiver configurations open. O leaves the eigensystem and phase winding policy open. B leaves a complete producer calibration and record protocol open. The master exposes those items in its configuration rather than filling them from general knowledge. [C, pp. 3–5][O, pp. 6–7][B, pp. 7–8]

An operation that does not need an absent parameter can still be defined. For example, literal orientation needs increments but not an axis; the residual needs both. The documented word/JK worked tick needs no external sensor protocol. This compositional validity is part of the total engine's interface, not an excuse to substitute a smaller panel for it.

33 Metrics and the cost of the integrated engine

Source descriptions attach several domain meanings to counts, curvature and timing. The master keeps numerical metrics typed by what they actually count in the chosen engine profile. [A, pp. 27–35]

Metric	Definition	Meaning
Word occupancy	$\text{popcount}(x)$	Set cells in this word, not area or mass by itself.
ASA/NA selection	$\sum \text{popcount}(a), \sum \text{popcount}(n)$	Pre-disposition support counts.
Boundary intersections	$\sum h$	Selected boundary bits, distinct from absorbed words.
Absorbed word count	$\sum \mathbf{1}[h > 0]$	Logical groups absorbed by the whole-word rule.
Offspring ratio	$ F_{g+1} / F_g $ when denominator is nonzero	Procedural lineage ratio for the selected profile.
OTAN2 residual	$e_{raw}, e_{line}, e_{directed}$	Profile/frame-specific orientation comparison.
Invariant coverage	Defined checks / requested checks	Must report UNDEFINED separately from PASS.
Compute duration	Explicitly delimited elapsed time	Backend/sample measurement with stated inclusions.

Log-radial cells need not have equal Euclidean area. A planar cell with radial endpoints r_i, r_{i+1} and angular width $\Delta\phi$ has area $\frac{1}{2}(r_{i+1}^2 - r_i^2)\Delta\phi$. A weighted-area report can use that measure; population count alone cannot be silently called volume.

For L_w logical words, the fixed-mask word pass has $O(L_w)$ work and $O(L_w)$ result storage. The constant work per word does not make a whole bank constant-time. A branching lane without pruning can have 2^g nodes at depth g . RK4 work scales with the number of evaluated states, substeps and field evaluations. Provenance scales with emitted record bytes.

An OTAN2 bank over N observations adds $O(N)$ evaluations and diagnostic records. The programmable lane adds work proportional to the performed transition prefixes and symbol representation. Device transfers, synchronization and storage are not deleted from the cost merely because a lookup has a fixed-width address.

No integrated GPU speedup is claimed from these counts. Texture/global and linear/Morton performance must be compared with the same input semantics, size, precision and timed region. The source's broad A*/grid comparisons remain source statements rather than new benchmark results for M1.

34 Verification performed for this master

This edition re-read the supplied PDFs and source package, reran the available mathematical component tests, and executed a new finite audit of the master word/JK/blend/OTAN2 assembly. The categories below are not conflated.

Check actually rerun	Result	Scope
Supplied I2 JavaScript core	30 groups; 109,154 assertions passed	OTAN2, invariants, ASA, blend, JK and original pure-annotation behavior.
Independent Python/Node comparison	1,218 snapshots; 6,078 comparisons passed	Angle/status results under both selected profiles.
New M36 assembly audit	17 groups; 20,873 assertions passed	Integrated word/JK tick, statuses, neutral recovery, isolation, replay and commit record.

The Python/Node comparison observed maximum absolute difference $8.881784197001252 \times 10^{-16}$ radians over its finite tested values, below its 10^{-12} tolerance. The JavaScript runtime used for these reruns was Node.js v22.16.0. These are local arithmetic/component results, not hardware throughput claims.

34.1 New assembly checks

The M36 audit evaluates the worked tick, deterministic replay without mutating its inputs, observation noninterference with core/JK state, undefined-OTAN2 handling, missing blend encoding, explicit fringe activation, exhaustive two-bit mask tuples, the full JK table, invalid-input rollback, time-interval rejection, ledger predecessor binding, Morton tile bijection, source/completed distinction unit conversion, and the declared lineage-to-word carrier including shared-cell identities.

Its exhaustive two-bit group covers all $4^6 = 4096$ combinations of the six small word operands and five assertions per combination. This explains much of the assertion total. The audit is a reduced abstract implementation of the M1 schedule using the supplied pure mathematical functions; it is not a newly compiled production engine.

34.2 Other evidence and remaining execution categories

The supplied I2 package contains a historical 19-check browser report. That UI test was inspected but was not rerun for this PDF-only consolidation. Earlier ASA/GPU test counts stated elsewhere in the conversation are not promoted into fresh M1 results. A complete assembled runtime build, GPU execution and device memory tests were not performed for this document.

The master equations and compatibility proofs remain independently readable. Runtime conformance is a separate implementation milestone; the new file is a completed master specification, not a claim that code was installed on the author's laptop.

35 Master assembly decisions and import mapping

ID	Decision	Effect
M01	One master version label	Title the integrated engine v3.6; retain v3.6.1 and I1/I2 as historical component labels rather than produce another addendum.
M02	Product state	Give genesis, words, lineage, JK, U, ledger and diagnostic records one explicit state structure.
M03	Snapshot dependency order	Resolve required types/configuration first, evaluate named components, publish only at commit.
M04	Declared chart profile	Preserve log normalization and expose quantization; distinguish inherited review values from source constants.
M05	Topology selector	Retain annulus and quotient realizations as named profiles; do not infer one from the other's masks.
M06	Separate word and lineage forms	Keep both literal OR/XOR splits and the separately introduced valid-child-address realization.
M07	Named increment origin	Record whether an OTAN2 sample came from supplied increments, an endpoint pair or a declared evolution segment.
M08	Explicit JK clocking	Evaluate from old state and supplied J/K; retain state without an event; add no hidden angle-to-JK policy.
M09	Master commit equations	Make ASA outputs, JK state and OTAN2/invariant records one next-state publication. Undefined observations and fatal inputs remain distinct.
M10	Explicit status aliases	Canonicalize I2 executable names while preserving I1 document spellings as import aliases.
M11	Whole-engine memory accounting	Add all enabled lanes and records to the word-kernel payload before applying resource checks.
M12	Versioned master envelope	Keep component semantics while separating new master serialization/hashes from older records.
M13	Retained U extension	Restate the earlier programmable construction and its logical-memory assumptions without attributing it to the OTAN2 source.
M14	Finite worked configuration	Choose a small local word/JK/observation tick to demonstrate and audit the full record/commit contract.
M15	Explicit lineage–word carrier	Quantize declared candidates, emit shared cell support, and admit registry entries by the corresponding final word bits. Keep IDs distinct and label this as a new adapter.

35.1 What is no longer external to the specification

The ASA word rule is given here; the JK recurrence is given here; OTAN2's two profiles, angular conventions and statuses are given here; the invariant bank is given here; and the master transition, record and memory contracts are given here. Prior documents support provenance, but readers need not leave this PDF to reconstruct those definitions.

Parameter fields left open by the original material remain visible. Consolidation changes organization and supplies declared assembly structure; it does not quietly invent the unanswered numerical or external-system details.

36 Source vocabulary and non-executable labels

Source label	Place in the integrated engine
BBV; wingnut / judo flip	Origin of the hinge and invariant-axis vocabulary; a selected transformation must still state what object it acts on.
V; womb; gestation	Proposed state becoming a retained trace; not the valid-bit mask or an automatically selected biological delay.
Ψ ; lower-case ϕ ; “dephined”	Typed state, phase and dictionary interpretation. The literal source spelling is retained.
Double UU; double dots; capsule	Identity/state binding and symbolic genesis; full digest and state are separately represented.
Hadamard blend / circled plus	Source label for the defined Boolean XOR profile; separate from AND intersection and numerical angle arithmetic.
ASA >0<; NA ASA	Paired aperture and post-ASA selection/lens interface; numerical profile identified separately.
North polar pinion; A–W–B–N	Frame/interval/reference record from C, with three distinct predicate meanings.
OTAN2; moving eigenvalue axis	Source ratio and supplied relative-axis subtraction, with an explicitly selected directed completion.
Inuit V-mirror; SSymmetry	Conditional reflection relation and frame/topology definitions.
Vantablack; Hollowland; Et. AI. VI.	Source sink/word-mark vocabulary; only a declared local mask/disposition participates in the computational state.
JK; Joker; Queen B / Royal Jelly	JK’s Boolean state role is retained. Narrative biological or authority labels do not define an additional arithmetic or execution interface.
Aperture / portal; Mirage / Ring	Paired index or presentation vocabulary. No external game integration or hidden privilege follows from the name.
Fly eye; ommatidia; optic lobe	Faceted input organization, edge-reduced producer port and mosaic record vocabulary.
OMT; ICU; R-value	Source domain labels; engine counts and parameters keep their explicit computational definitions.
PRISM, S2, satellite, neural or weapon names	Source context only. No external feed, individual-target matcher, privileged connection or actuator is created by the master contract.

This glossary preserves the author’s corpus labels without allowing repeated names to overwrite the typed operators. It also distinguishes a textual metaphor from a callable function: code or pseudocode requires an explicitly defined input/output contract. The source’s claims of active external-system status are not evidence that such systems are connected to a PDF or local state record.

A Cross-document source map

Source	PDF pages	Material carried into the master	Master sections
R0 / A	R0 6–15; A 6–14	BBV, source alphabet, register split, lower-case phase and gestation	1, 7, 8
A	16–18; 75–78	Double UU, identity, corrected display genesis	4
A	19–24; 30–38	RK4, angular collapse, divergence, predicate words and sink	5–9, 13
A	53–55	JK set/toggle source context	10
A	75–85	ASA, NA after ASA, portal pairs	14, 6
A	85–91	Bit swizzle and fly-eye addition	23, 25
B	4–8	Compound-eye framing, facets, edge reduction and rate target	25
C	1–2	A–W–B–N, circle-plus source expression	11
C	2–5	Three Boolean meanings and hollow-center refinement	12
O	1–5	Atan2 background, phase helpers, spelling correction	15, 16
O	6–7	Literal ratio, moving pinion/axis, relative subtraction	15–18
P31	4–12	Retained ASA core, fringe option, namespacing and recovery	11–14, 31
J361	4–13	JK contract, two OTAN2 profiles, phase/frame/status conventions	10, 15–22
I2	4–13; supplied code	Six invariant laws, pure functions and numeric tolerances	17, 18, 34
M1	Current assembly	Product state, explicit tick and commit, worked integration	3, 19–22, 29

Not a concatenation. The original pages are not stapled together as the master body. Their definitions and distinctions have been consolidated into the single preceding sequence. Page citations preserve the ability to compare a source expression with the inherited and newly declared formalization that uses it.

Versioned interpretation. When a source uses one word for different operations, the master names the alternatives and cites their locations. When an earlier integration adds a convention, the master attributes that convention to that integration rather than to the original source.

B Evidence records and source fingerprints

This edition was prepared from actual local PDF files and the supplied I2 source ZIP. The following compact fingerprints identify the principal inspected records. All hashes are SHA-256; the full values are printed to avoid relying only on similar filenames. **A — WHITE KING corpus.** 91 pages

ba3dbe6f925c68ac0f237184b1a4f7770c4fe1d5c84b54bcb5a108dc37ec2ce3

B — Fly-eye addendum. 8 pages

1a91d9ce2af8c1ebdfc460913438cf127f1bde8be67c5a01c410976b5d7299ad

C — Blend operator source. 5 pages

7c8e25b7269dc96b96e5e71afabb8d0019e905975f9959569ee4488b2c036ac8

O — OTAN2 source. 7 pages

90d8b3abadccce971917ea68e4f9be7b9d8d4a52ad7b700066649eb3ebc29732

P31 — ASA 3.1 blend integration. 20 pages

3b7d2c839f29ca4a30f86276081ac59bd69d6294e0e9fba06988d0944f38247a

J361 — JK v3.6.1 integration I1. 23 pages

7c7a216192121e83f27cc1ab1d0e2dd6d390f176486bbab8b74c1749b8cf6359

I2 — v3.6 invariant specification. 16 pages

da4be2ca1e6784479026a443f9544de790f1bf53ebf055064137ad0c99d1acc7

I2-Z — Supplied invariant source package. ZIP source package

3ca5287b5249ea2f44495702b347e22647dca51f398c0fe6d66d9dd2d3adadfe

Executed audit records. Section 34 reports the actual results and scope of these retained preparation records:

node_tests.json

cross_reference.json

master_contract_tests.json

master_tick_example.json

Their information is incorporated into this PDF. No separate executable or ZIP was requested for this consolidated delivery.

Reproducibility boundary. The supplied I2 package contains the component tests and independent Python evaluator. M1's new audit checks the displayed master assembly, not a patched binary. A future code release can implement this full contract and report additional build/device evidence without altering the mathematical/source distinctions here.

C Primary hardware references and publication record

Hardware facts were checked separately from the conceptual sources on 13 September 2026. They support the backend chapter; they are not alternative definitions of OTAN2.

- H1 NVIDIA, [Compare GeForce RTX Laptops](#). RTX 5070 Ti Laptop memory configuration: 12 GB GDDR7.
- H2 NVIDIA, [CUDA GPU Compute Capability](#). RTX 5070 Ti family listed at 12.0.
- H3 NVIDIA, [CUDA 12.8 Features](#). Historical introduction of compiler support for SM_120; this is a minimum-target reference, not a claim about the newest toolkit.
- H4 NVIDIA, [CUDA Runtime API: Texture Object Management](#). Resource descriptors, element reads, linear-resource alignment/extent and sampling-property behavior.
- H5 Microsoft, [User Mode and Kernel Mode](#). Host application/driver privilege boundaries.

Publication record

Title: atomOS v3.6 — Total Integrated Engine Specification.

Author attribution: Tom Klootwijk.

Author-supplied identifier/date: NL200678942 / 10-07-1990.

Master edition: M1, 13 September 2026.

Scope: a consolidated, self-contained mathematical and software-interface specification joining the documented core, JK, ASA/NA, blend, OTAN2, invariants, topology, memory and provenance.

This edition was prepared with AI assistance from the supplied corpus and prior artifacts. New M1 assembly choices and finite audits are identified as such. A reader can distinguish source statements, inherited conventions, deductions, executed mathematical checks and future implementation conformance without consulting an implicit or invented release history.

atomOS v3.6

The engine is the integrated state transition and its complete record: identity, state evolution, aperture, JK, OTAN2, invariants, memory and provenance. The control panel is one view of that record; it is no longer the scope of the deliverable.